Diksha Shekhawat

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EDUCATION

Indian Institute of Technology Kanpur — Kanpur
Semester Exchange Student — Semester IV

JK Lakshmipat University — Jaipur
B.Tech in Computer Science & Engineering — CGPA: 9.016/10 (Sem I: 8.45 | Sem II: 9.50 | Sem III: 9.095)

St. Teresa's School — *Class XII (CBSE)* — 90.2% | 2023 | Jaipur

St. Teresa's School — *Class X (CBSE)* — 94.4% | 2021 | Jaipur

WORK EXPERIENCE

Vicharanashala Lab for Education Design (IIT Ropar) — 15 May 2026 - 16 July 2026

- Engineered a **real-time multimodal monitoring pipeline** integrating webcam video and microphone audio with lecture streaming in a React-based web application.
- Implemented **face detection, facial landmark tracking, head-pose estimation, face-count monitoring, and attention-loss detection** using **MediaPipe Computer Vision**.
- Built browser-side **Voice Activity Detection (VAD)** using **Web Audio APIs** to identify speech events during monitored learning sessions.
- Developed an **event-fusion layer** combining audio and visual signals into timestamped behavioral events for instructor review and learning analytics.
- Designed backend APIs and persistence using **Node.js, Express.js, and MongoDB** for session telemetry, activity logs, and detected events.
- Implemented analytics dashboards providing **lecture progress, attendance insights, and chronological behavioral-event timelines**.

Oeja Swasthasetu — July 2026 - Ongoing

- Developed backend architecture for **OejaRekha** using **Node.js, Express.js, MongoDB and Mongoose**, modelling users, Care Partner profiles, tasks, sessions, verification records, incidents and disputes.
- Implemented **phone + OTP authentication with JWT access/refresh tokens**, including secure OTP hashing, expiry handling and role-based backend access for the Care Partner workflow.
- Built **geospatial service-radius enforcement** using MongoDB **2dsphere indexing**, enabling proximity-based Care Partner discovery and server-side validation of the 20 km assignment constraint.
- Implemented backend **state machines and validation rules** for Care Partner onboarding, task lifecycle, session check-in/check-out, incidents and dispute workflows.
- Developed scheduled background jobs with **Node.js/Node-Cron** for OTP expiry cleanup, verification-badge expiration, automated escrow-state release and recording-retention workflows.
- Designed an append-only **EventLog** architecture for timestamped platform events, providing an auditable evidence trail for operational, incident and dispute workflows.

ACADEMIC PROJECTS

Database Engine — **C++, B+ Tree, SQL, File I/O, Concurrency**

- Engineered a **disk-based relational database engine** supporting SQL parsing, query execution, and persistent table storage.
- Implemented **B+ tree indexing, buffer management, page-based storage, and record serialization** for efficient data access.
- Built a **query execution pipeline** supporting filtering, projection, insertion, deletion, and indexed lookups.
- Added **transaction/concurrency control and recovery mechanisms** to maintain consistency across concurrent operations.

Network Intrusion Detection System (NIDS) — **Python, Scapy, TCP/IP, Machine Learning, Linux**

- Engineered a **real-time packet inspection pipeline** for network traffic capture, TCP/UDP flow reconstruction, feature extraction, and intrusion detection.
- Implemented **signature/rule-based detection** for port scans, SYN floods, ICMP floods, and anomalous connection behavior.
- Developed **flow-level statistical features** including packet/byte rates, connection duration, TCP flags, protocol distribution, and destination-port entropy.
- Integrated an **ML-based anomaly classifier** and evaluated detection using **precision, recall, F1-score, and false-positive rate**.
- Built **real-time alerting and traffic visualization** with attack classification, source attribution, timestamps, and severity.

SKILLS

Programming Languages: Java, C, C++, Python
Frontend: HTML, CSS, Bootstrap, JavaScript, React.js, Three.js
Backend & Database: Node.js, Express.js, MongoDB
Tools & Technologies: Git, GitHub, Docker, Kubernetes

ACHIEVEMENTS

- **Swasth-a-thon | Hackathon: Third position**, built an app with a virtual mascot to remind the labour force from time to time for screen breaks, hydration breaks, and act as an AI assistant.

POSITION OF RESPONSIBILITY

- **Cohort Leader – Freshers' Orientation** | *JK Lakshmipat University* — 2025
- **Stage and Venue Core – Sabrang (Cultural Fest of JKLU)** | *JK Lakshmipat University* — 2026
- **General Secretary Academic Affair** | *Student Council* | *JK Lakshmipat University* — 2026

CERTIFICATIONS

- Introduction to backend development, Introduction to frontend development – Meta
- AI for everyone – DeepLearning.AI
- Machine Learning for Introduction for everyone – IBM